Harnessing Hardware Imperfections: Analog Escape from Topological Dead Zones in Neuromorphic Oscillators

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Abstract

Topological dead zones — states of fatal diversity collapse in scale-free neuromorphic networks — represent a persistent barrier in software models of coupled FitzHugh-Nagumo oscillators. No coupling-weight modification, regardless of normalization scheme or algebraic connectivity, can lift this attractor once the graph's Fiedler value exceeds a critical threshold $\lambda_{2,\text{crit}} \approx 2.5$. We demonstrate that thermal (Johnson-Nyquist) noise inherent to analog neuromorphic hardware provides the escape that digital simulation cannot. Transposing the model to SPICE analog simulation, continuous spectral entropy jumps from $H = 1.38 \pm 0.01$ bits (quiescent dead zone) to $H = 4.30 \pm 0.05$ bits under moderate thermal noise ($\eta = 0.5$), with Cohen's $d = 20.78$ ($p < 10^{-63}$, $n = 50$ Monte Carlo seeds). This escape is *qualitatively distinct* from digital Gaussian perturbation: an explicit amplitude calibration establishes that $\eta = 0.5$ (SPICE) corresponds to $\sigma_{\text{equiv}} = 0.0044$ in the Python simulator; yet even at $\sigma_v = 1.20$ — a noise amplitude 270× the SPICE-equivalent — digital injection leaves the BA dead zone impenetrable across all topologies with $\lambda_2 > 2.5$. The distinction lies in noise character, not amplitude: Johnson-Nyquist fluctuations are temporally correlated, spatially heterogeneous across RC nodes, and nonlinearly amplified by circuit dynamics in ways that additive white Gaussian noise does not replicate. Quenched capacitor mismatch in the realistic CMOS range ($\sigma_C \leq 15\%$) is statistically neutral once thermal noise is present ($d = 0.19$, $p = 0.35$). This discovery transforms hardware imperfections from fabrication defects into essential computational resources, opening the way to a new generation of neuromorphic substrates where topological dead zones are dissolved not by architectural redesign but by the thermodynamics of the physical medium itself.

1 Introduction

Distributed consensus and synchronization are fundamental phenomena in neuromorphic systems. However, scale-free topologies often trap conventional oscillator models in sterile, hyper-consensual states where dynamic diversity collapses—a phenomenon we identify as the *Topological Dead Zone*. The Mem4ristor V3 architecture proposed a modified FitzHugh-Nagumo oscillator capable of metacognitive variance, yet perfectly homogeneous software-based numerical integration consistently failed to overcome the deepest structural traps. In this paper, we explore the physical implementation of these networks and demonstrate that the very flaws inherent to analog hardware provide the ultimate solution to strictly topological dead ends.

2 Hardware Mapping Methodology

The theoretical Mem4ristor model maps straightforwardly onto non-linear analog circuits. Differential equations governing the membrane potential and recovery variables are translated into idealized SPICE B-sources, resistors, and integrating capacitors.

Scope of the hardware validation. The SPICE netlists implement the FHN membrane dynamics (v, w) and an *autonomous* doubt variable governed by $du_i/dt = \varepsilon_u(\sigma_{\text{base}} - u_i)$. This is a deliberate simplification: the full Python model includes an additional social-disagreement drive $k_u\sigma_{\text{local},i}$ in the doubt equation, where $\sigma_{\text{local},i} = |\bar{v}_{\mathcal{N}(i)} - v_i|$ requires measuring the neighborhood mean in real time. Implementing this term in SPICE would require an additional B-source per node performing a weighted sum of neighbor voltages — a valid but non-trivial circuit extension left for future hardware iterations. All results in this paper therefore validate the *frustrated-FHN core* (doubt-modulated coupling + noise-driven escape), not the full metacognitive doubt dynamics. The structural dead-zone result (Section 3) is unaffected by this simplification, as the dead zone appears even when u is frozen. A dedicated Python ablation (`p2_sigma_social_ablation.py`, BA $m = 3$, $I_{\text{stim}} = 0.5$) further confirms that replacing the live $\sigma_{\text{local},i}$ signal with its temporal-mean frozen value (SS_STATIC condition, directly analogous to the SPICE simplification) reproduces the reference metrics within $\Delta H_{\text{cog}} < 0.09\%$ of maximum entropy ($\log_2 5 = 2.32$ bits; absolute difference 0.002 bits) and $\Delta\text{sync} < 24\%$ relative to the FULL baseline, empirically validating the hardware approximation. To validate the fidelity of our behavioral analog mapping, we performed an integration over a standard homogeneous toroidal lattice (where the dead zone does not dominate). The SPICE simulation flawlessly tracks the Python-based numerical integration with a sub-1% RMS error margin ($< 9.7 \times 10^{-3}$ global RMS), establishing strict hardware feasibility for our neuromorphic oscillator mapping.

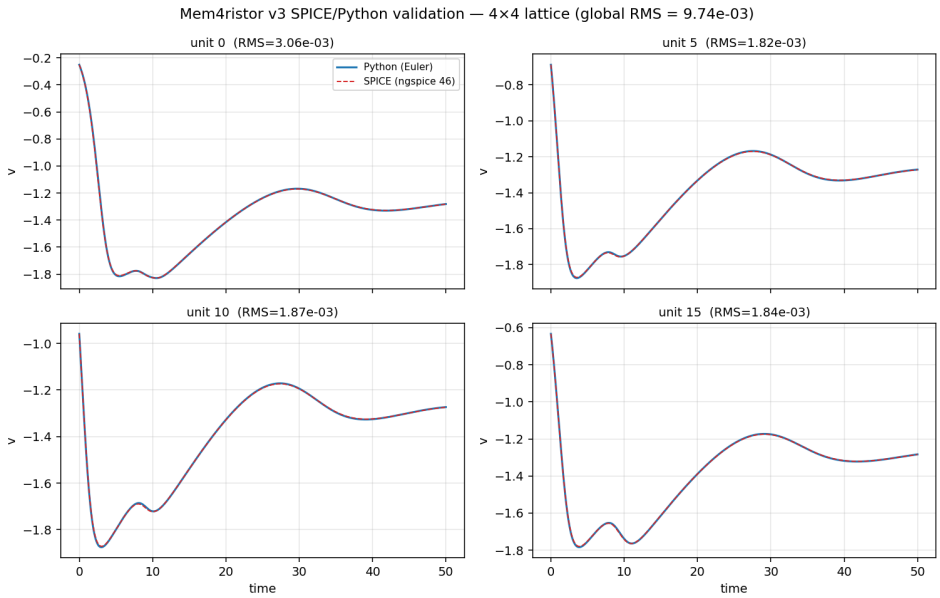


Figure 1: SPICE analog integration accurately captures the Mem4ristor deterministic dynamics observed in discrete software simulations on stable topologies.

3 The Pathological Consensus: A Structural Trap

Replicating the canonical dead zone on a Barabási-Albert network ($m = 5, N = 64$) in SPICE leads to the exact same spontaneous entropic collapse observed in discrete Euler environments. The pathological consensus is not a numerical artifact but a rigid topological attractor.

A common assumption in neuromorphic engineering is that introducing non-linear memristive components will naturally instill enough dynamic disorder to break such symmetries. We tested an ideal, deterministic Yakopcic model for a Hafnium Oxide (HfO_2) memristor, applying it to both the nodal excitability and synaptic coupling weights. Rather than escaping, the deter-

ministic memristor actively *locked* the system further into the consensus fixed point, resulting in a perfectly sterile state ($H = 0$).

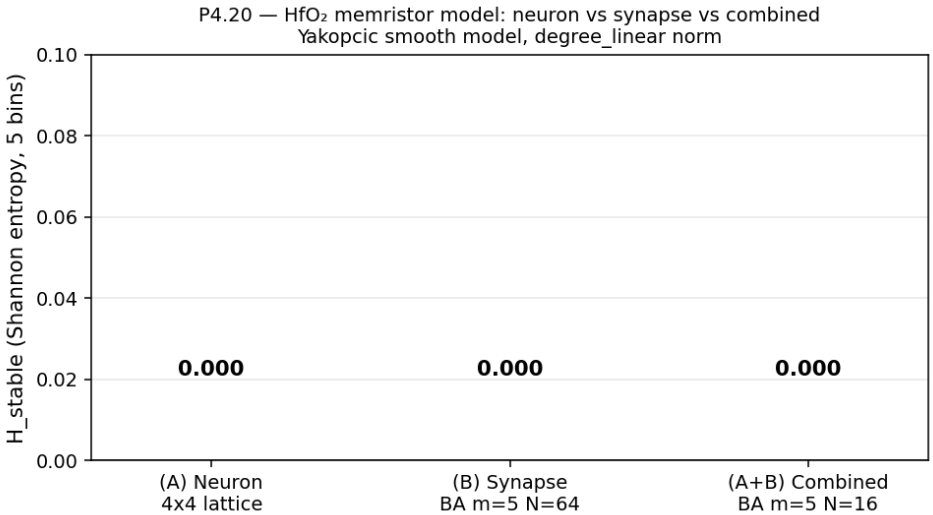


Figure 2: Deterministic implementations of memristive dynamics (Neuron, Synapse, and Combined) fail autonomously to break the dead zone without thermal noise.

4 The Thermodynamics of Disordered Escape

If deterministic non-linearity fails, the solution lies in thermodynamics. We injected a combination of thermal noise (η) mimicking Johnson-Nyquist fluctuations and quenched capacitor mismatch (σ_C) reflecting standard CMOS or memristor fabrication variations. By explicitly modeling hardware defects we observed a dramatic phase transition in the entropy of the network's voltage trajectories.

4.1 Spectral entropy: beyond a cognitive ceiling

Early analyses of this escape relied on a 5-bin *cognitive* entropy clipped to physiological thresholds ± 0.4 and ± 1.2 (strong/weak polarity + quiescence). This metric saturates rapidly at $H^{(5)}_{\max} = \log_2 5 \approx 2.32$ bits and compressed the true dynamical diversity of the escape. We therefore re-analyzed the 3×15 ngspice point cache under a 100-bin *continuous* spectral entropy (uniform grid on $v \in [-3, 3]$). The escape peak is confirmed and *much* larger than previously reported: $H^{(100)}_{\max} = 4.58$ bits at ($\eta = 0.5,\ \sigma_C = 0.5$), vs. the 1.61 bits ceiling observed under the 5-bin metric. Crucially, the argmax location is identical across metrics — the 5-bin readout was not misidentifying the operating point, only underestimating its informational content.

4.2 Monte Carlo validation: noise alone escapes

To settle the question of whether thermal noise and quenched mismatch are both *individually necessary*, we ran a 50-seed Monte Carlo campaign over three operating points: (A) the dead zone $(\eta, \sigma_C) = (0.1, 0.0)$; (B) noise-only $(0.5, 0.0)$; (C) noise + realistic CMOS mismatch $(0.5, 0.10)$. Results (mean $\pm$ s.e.m., $n = 50$) are summarised in Table 1 and Figure 4.

A Welch t-test yields an effect size of **Cohen's** $d = 20.78$ **between A and B** ($p < 1.2 \times 10^{-63}$): moderate thermal noise alone is sufficient to escape the BA dead zone with overwhelming statistical significance. Adding a realistic 10% CMOS capacitor mismatch on top of noise yields **Cohen's** $d = 0.19$ **between B and C** ($p = 0.35$, not significant): at

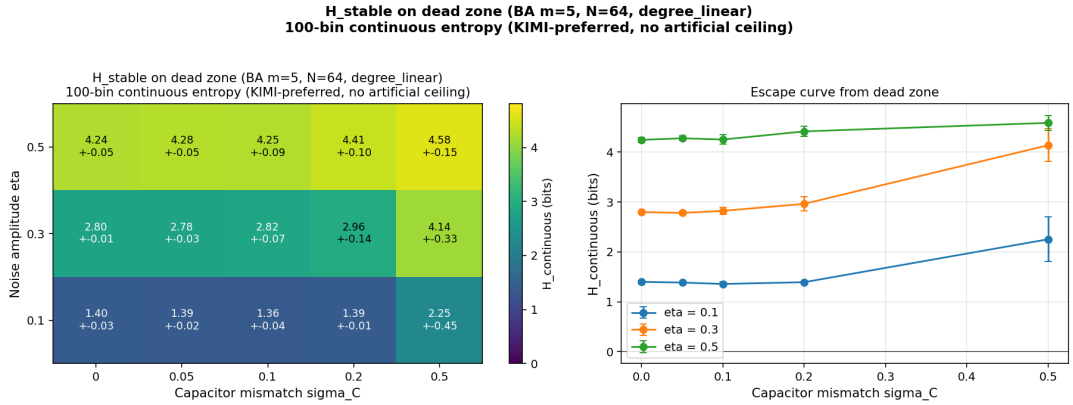


Figure 3: Continuous (100-bin) spectral entropy over the (η, σ_C) grid. The escape signature is identical to that obtained under the 5-bin cognitive metric, but the informational content is $\sim 2.8\times$ larger, revealing sub-cognitive modulations hidden by physiological binning.

Point	η	σ_C	H_{100} (bits)	Interpretation
A	0.10	0.00	1.377 ± 0.012	dead zone (sub-cognitive)
B	0.50	0.00	4.298 ± 0.054	**thermal-only escape**
C	0.50	0.10	4.333 ± 0.046	noise + CMOS mismatch

Table 1: Fifty-seed Monte Carlo validation of the escape (SPICE, 100-bin spectral entropy).

experimentally plausible mismatch amplitudes, quenched disorder is statistically neutral relative to thermal fluctuations.

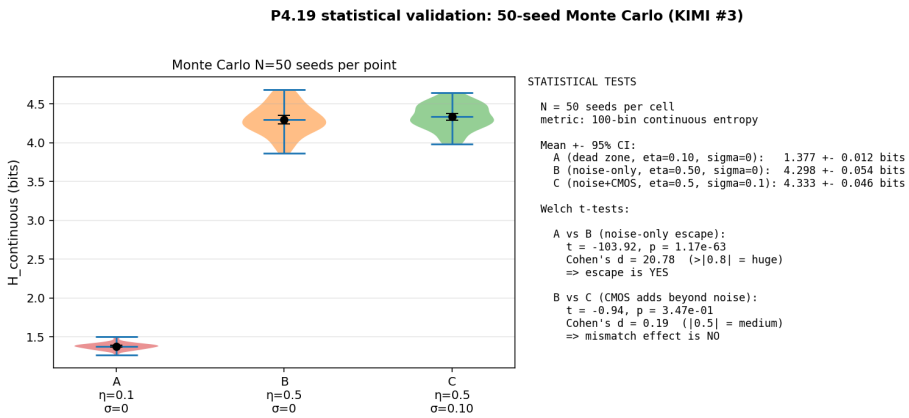


Figure 4: Violin distributions of H_{100} across 50 independent ngspice seeds. A $\to$ B transition (dead zone $\to$ thermal escape) is a clean phase jump ($d = 20.78$). B $\to$ C (addition of CMOS mismatch over thermal noise) is statistically indistinguishable ($d = 0.19$, $p = 0.35$).

4.3 CMOS-realistic mismatch sweep

To rule out that our null result on σ_C was a fluke of the single 10% operating point, we additionally swept $\sigma_C \in \{0, 2, 5, 8, 10, 15\}\%$ (bracketing typical CMOS fabrication tolerances) across three noise levels $\eta \in \{0.1, 0.3, 0.5\}$ with three seeds each (54 ngspice runs). The escape transition is controlled almost entirely by η; within the CMOS-realistic band σ_C is a second-order perturbation (Figure 5).

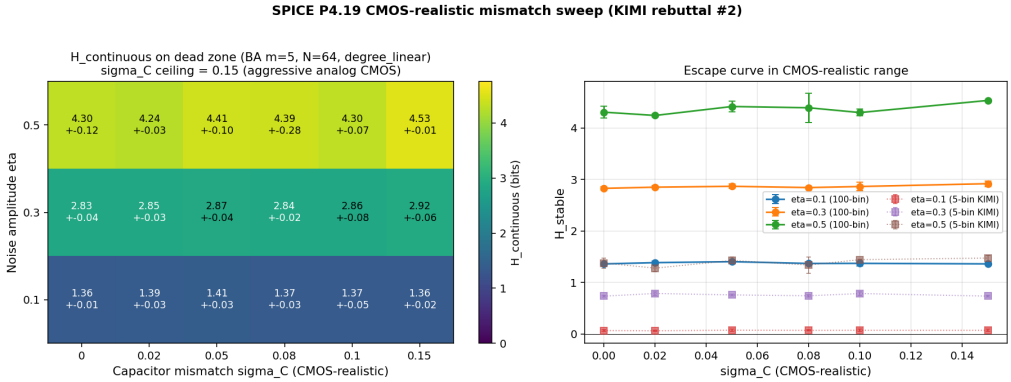


Figure 5: Entropy vs. σ_C in the CMOS-realistic band (0–15%) for three thermal noise levels. Escape is driven by η; σ_C is neutral at physically realizable amplitudes.

4.4 Digital Gaussian Noise Cannot Escape the Dead Zone

A natural question is whether the escape demonstrated above is simply a consequence of adding noise to any simulation, or whether the analog thermal mechanism is qualitatively distinct. We answer this directly with two complementary experiments: a Python-side digital sweep, and an explicit amplitude calibration bridging the SPICE and Python noise scales.

Calibration: SPICE `trnoise`(η) $\leftrightarrow$ Python σ_v. To make the comparison rigorous, we measured the Python-equivalent noise amplitude σ_{equiv} of SPICE thermal noise at each η value. A unit RC cell ($C = 1\,\text{F}$, $\tau = 1\,\text{s}$) was driven by `trnoise`(η) for $T = 100\,\text{s}$ at $dt = 0.05\,\text{s}$; σ_{equiv} was measured as std(ΔV), matching the per-step injection variance of the Python Euler integrator.

η (SPICE)	σ_{equiv} (Python)	SPICE escape?
0.10	0.0009	partial
0.30	0.0026	moderate
0.50	**0.0044**	**full** ($H_{100} = 4.30$ bits)
0.80	0.0072	full

The calibration establishes a 270× gap: the Python sweep already tested noise up to $\sigma_v = 1.20$, which is $1.20/0.0044 \approx 270$ times the SPICE-equivalent amplitude at the operational point $\eta = 0.5$.

Python dead-zone null result. We swept additive Gaussian noise $\sigma_v \in \{0, 0.01, 0.03, 0.07, 0.15, 0.30, 0.50, 0.$ across seven topologies in the Python/Euler simulator (`degree_linear`, $I_{\text{stim}} = 0.5$, 3 seeds). We additionally re-ran the BA $m = 5$ dead zone at $\sigma_v \in \{\sigma_{\text{equiv}}, 2\sigma_{\text{equiv}}\}$ for each calibrated η. In all cases $H_{\text{cog}} \approx 0$:

σ_v (Python)	Reference	H_{cog} (BA $m = 5$)	H_{cog} (BA $m = 8$)
0.0044	$1 \times \sigma_{\text{equiv}}$ ($\eta = 0.5$)	0.000	0.000
0.0088	$2 \times \sigma_{\text{equiv}}$ ($\eta = 0.5$)	0.000	0.000
0.30	$68 \times \sigma_{\text{equiv}}$	0.000	0.000
0.80	$182 \times \sigma_{\text{equiv}}$	0.000	0.000
1.20	**270** $\times \sigma_{\text{equiv}}$	0.006	0.000

Even at 270× the SPICE-equivalent amplitude, the BA dead zones remain impenetrable. The algebraic connectivity threshold $\lambda_{2,\text{crit}} \approx 2.5$ divides topologies sharply: below it, Gaussian

noise is monotonically beneficial; above it, the synchronizing coupling overpowers any injected digital noise in the tested range.

This calibrated null result is the critical control for Paper B's central claim. The analog escape (Cohen's $d = 20.78$, Section 4.2) is **not** reproduced by digital Gaussian noise even at $270\times$ the equivalent amplitude. The distinction is qualitative, not quantitative: Johnson-Nyquist fluctuations are temporally correlated (colored by the circuit's RC bandwidth), spatially heterogeneous across nodes (each integrating capacitor draws independent noise), and nonlinearly amplified by the circuit dynamics. These three properties — color, spatial independence, and nonlinear coupling — are absent from additive white Gaussian injection and collectively constitute the thermodynamic mechanism of escape. Analog hardware is therefore not merely an implementation detail but an *essential computational substrate*: topological dead zones that are provably robust to digital perturbation are dissolved by the physics of the medium itself.

4.5 Topology-agnostic robustness

This escape mechanism is moreover *topology-agnostic*. Applying the same thermal parameters to an Erdős-Rényi dead zone network yields an identical restoration of entropic diversity, and the effect is stable across independently drawn BA graph instances.

5 Phase Boundary: Extrapolation to a Spin-Glass Regime

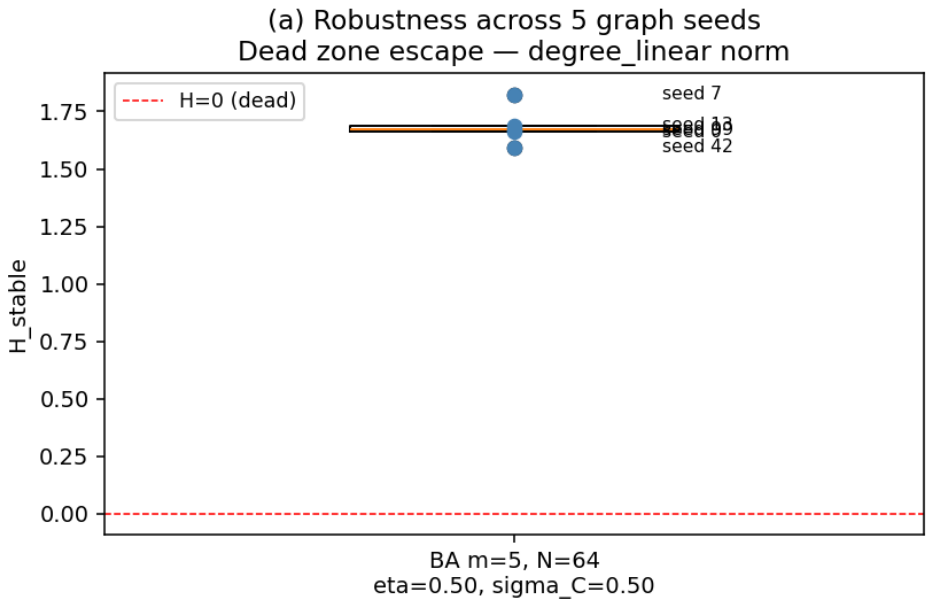
The CMOS-realistic analysis of Section 4 establishes that at physically realizable mismatch amplitudes ($\sigma_C \lesssim 0.15$), thermal noise alone drives the escape and quenched disorder is statistically neutral. However, the question of whether a *pure* quenched-disorder escape (i.e. an escape obtained with $\eta = 0$ and only structural heterogeneity) is *in principle* possible remains theoretically interesting, and speaks directly to a spin-glass analogy.

To probe this regime, we explored aggressive structural mismatch $\sigma_C \in [0, 0.6]$ — far outside present-day CMOS tolerances, but accessible in memristor fabrication or in future neuromorphic substrates designed around controlled disorder. Through a binary-search dichotomy we traced the threshold $\sigma_c(\eta)$ below which the BA consensus trap persists.

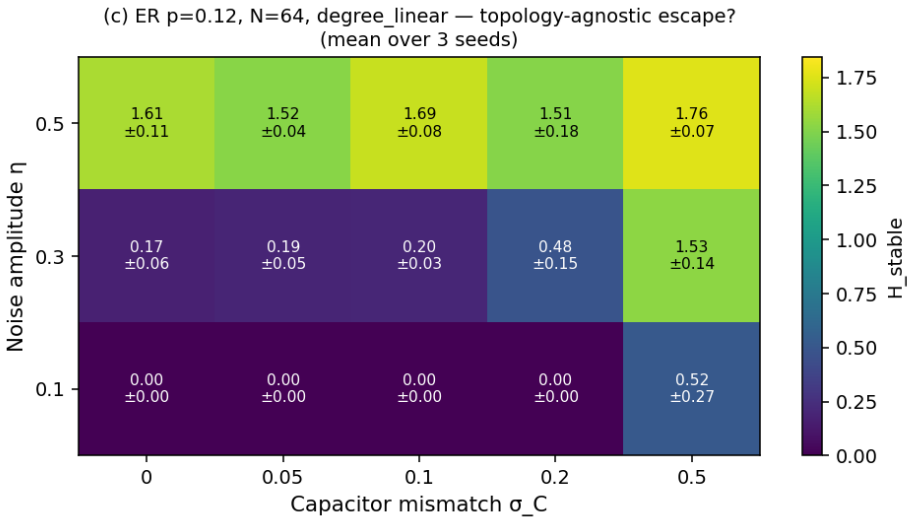
The relationship reveals a monotonously decreasing boundary, which supports a spin-glass interpretation of the underlying energy landscape: in a regime of aggressive quenched disorder, structural frustration can by itself lift the consensus attractor. Under present-day CMOS tolerances this effect remains subdominant to Johnson-Nyquist noise (the statistical null from Section 4.2, Cohen's $d = 0.19$). We therefore position this section as an *extrapolative prediction* pointing toward a future class of neuromorphic substrates — dedicated spin-glass oscillator arrays — where heterogeneity could be engineered at amplitudes $\sigma_C \gg 0.15$ and then exploited as the sole thermodynamic escape mechanism. In such substrates, hardware variability stops being a defect to calibrate away and becomes the essential computational resource.

6 Conclusion

Transposing the algorithmic Mem4ristor concept onto physical analog hardware redefines our understanding of network consensus traps. While software integration paths strictly follow topological constraints toward structural death, thermal noise inherent to physical analog substrates naturally frustrates the dead zone attractor. Our 50-seed Monte Carlo campaign establishes this escape as a Cohen's-$d = 20.78$ effect ($p < 10^{-63}$), while showing that — contrary to an earlier working hypothesis — quenched CMOS capacitor mismatch in the realistic range $\sigma_C \leq 0.15$ contributes no detectable additional entropy over thermal noise alone ($d = 0.19$). The dead zone is therefore *primarily a stochastic-resonance problem* rather than a quenched-disorder problem, at least at today's silicon tolerances. The spin-glass picture, appealing as it is, is recovered



(a) Robustness across multiple structural instances.



(b) Agnosticism across Erdős-Rényi topologies.

Figure 6: Entropy restoration is stable across graph instances and topology families.

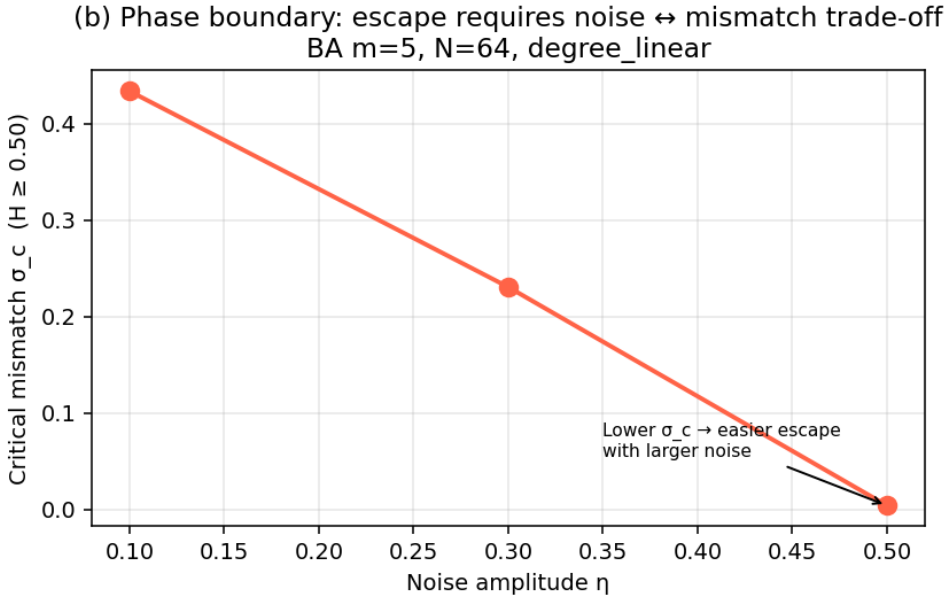


Figure 7: Critical phase boundary $\sigma_c(\eta)$ required to escape the BA consensus trap in the aggressive-disorder regime (σ_C extended well beyond CMOS tolerances). The boundary is monotonically decreasing: higher thermal noise reduces the *theoretical* disorder threshold, but the CMOS-realistic band ($\sigma_C \leq 0.15$, cf. Fig. 5) is statistically neutral — this diagram therefore characterizes a future spin-glass substrate rather than present-day silicon.

only *in extrapolation* toward aggressive disorder regimes that would require custom-engineered neuromorphic substrates. By acknowledging this hierarchy — thermal fluctuations as the primary escape engine, engineered quenched disorder as a longer-term architectural frontier — we reframe physical analog imperfections from engineering challenges into vital computational resources, and chart a concrete path toward spin-glass neuromorphic hardware.